Lab 7: Fuzzing for Input Validation Bugs

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SS1 (Step 2): Running the Buggy Application (python3 car_insurance_premium_parser.py)

```
nimayballal@MacBook-Pro Lab 7 % python3 car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;800
Traceback (most recent call last):
  File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 39, in <module>
    main()
  File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 32, in main
    print("Total claimed:", total_claim_amount(value))
                            ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    return sum(int(amount) for amount in text.split(';'))
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 16, in <genexpr>
    return sum(int(amount) for amount in text.split(';'))
               ^^^^^^^^^^
ValueError: invalid literal for int() with base 10: ''
nimayballal@MacBook-Pro Lab 7 %
```

Observation: Entering '12000;;800' into total_claim_amount() crashes the program immediately with ValueError: invalid literal for int() with base 10: '' because split(';') produces an empty token between consecutive semicolons.

SS3 (Step 5): Running the Fixed Application (python3 car_insurance_premium_parser.py)

```
nimayballal@MacBook-Pro Lab 7 % python3 car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ---
Normalized registration: Invalid input

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;;800
Total claimed: Invalid input

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14-08-1999
Driver age in 2026: Invalid input
nimayballal@MacBook-Pro Lab 7 % python3 car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 1234
Normalized registration: KA01AB1234

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;4500;800
Total claimed: 17300

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/08/1999
Driver age in 2026: 27
nimayballal@MacBook-Pro Lab 7 %
```

Observation: After hardening all three functions, malformed inputs ('---', '12000;;800', '14-08-1999') safely return None and print 'Invalid input' without crashing, while valid inputs ('ka 01 ab 1234', '12000;4500;800', '14/08/1999') return KA01AB1234, 17300, and 27.

SS2 (Step 4): Running Hypothesis Fuzz Tests on Buggy Code (Failing)



```
nimayballal@MacBook-Pro Lab 7 % pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.12.10, pytest-8.2.0, pluggy-1.7.0
rootdir: /Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7
plugins: hypothesis-6.170.14, anyio-4.10.0
collected 6 items

test_car_insurance_premium_parser.py FFF..F                              [100%]

=================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________
test_car_insurance_premium_parser.py:12: in test_normalize_registration_no_crash
    normalize_registration(s)
car_insurance_premium_parser.py:6: in normalize_registration
    plate = plate.strip().upper()
            ^^^^^^^^^^
E   AttributeError: 'NoneType' object has no attribute 'strip'

During handling of the above exception, another exception occurred:
test_car_insurance_premium_parser.py:10: in test_normalize_registration_no_crash
    def test_normalize_registration_no_crash(s):
                   ^^^
test_car_insurance_premium_parser.py:14: in test_normalize_registration_no_crash
    assert False, f"normalize_registration crashed with: {e!r}"
E   AssertionError: normalize_registration crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
E   assert False
E   Falling test case: test_normalize_registration_no_crash(
E       s=None,
E   )
_____________________ test_total_claim_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 18, in test_total_claim_amount_no_crash
  |     def test_total_claim_amount_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 20, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 16, in <genexpr>
    |     return sum(int(amount) for amount in text.split(';'))
    |                ^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 22, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e!r}"
    | AssertionError: total_claim_amount crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Falling test case: test_total_claim_amount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 20, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |                                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 22, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e!r}"
    | AssertionError: total_claim_amount crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Falling test case: test_total_claim_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________ test_driver_age_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 26, in test_driver_age_no_crash
  |     def test_driver_age_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 28, in test_driver_age_no_crash
    |     driver_age(s)
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 30, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e!r}"
    | AssertionError: driver_age crashed with: IndexError('list index out of range')
    | assert False
    | Falling test case: test_driver_age_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 28, in test_driver_age_no_crash
    |     driver_age(s)
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py", line 30, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e!r}"
    | AssertionError: driver_age crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Falling test case: test_driver_age_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________ test_driver_age_valid_and_future_dates _____________________
test_car_insurance_premium_parser.py:53: in test_driver_age_valid_and_future_dates
    st.integers(1, 28),
            ^^^
test_car_insurance_premium_parser.py:63: in test_driver_age_valid_and_future_dates
    assert result is None
E   assert -1 is None
E   Falling test case: test_driver_age_valid_and_future_dates(
E       # The test always failed when commented parts were varied together.
E       day=1,  # or any other generated value
E       month=1,  # or any other generated value
E       year=2027,
E   )
E   Explanation:
E       These lines were always and only run by failing test cases:
E       /Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7/test_car_insurance_premium_parser.py:63
=========================== short test summary info ============================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - Excep...
FAILED test_car_insurance_premium_parser.py::test_driver_age_valid_and_future_dates
========================= 4 failed, 2 passed in 0.30s ==========================
nimayballal@MacBook-Pro Lab 7 %
```

Observation: 4 failed, 2 passed. Hypothesis automatically falsified all three buggy functions: normalize_registration(None) raised AttributeError; total_claim_amount('') raised ValueError and total_claim_amount(None) raised AttributeError; driver_age('') raised IndexError, driver_age('//') raised ValueError, driver_age(None) raised AttributeError, and future dates like year=2027 returned negative age (-1).

SS4 (Step 6): Re-Running Hypothesis Fuzz Tests on Fixed Code (Passing)

```
nimayballal@MacBook-Pro Lab 7 % pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform darwin -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/nimayballal/Desktop/submissions/SE/Lab5/repo/Lab 7
plugins: hypothesis-6.170.14, anyio-4.10.0
collected 6 items

test_car_insurance_premium_parser.py ......                              [100%]

============================== 6 passed in 0.19s ===============================
nimayballal@MacBook-Pro Lab 7 %
```

Bugs Found by Hypothesis Fuzzing and Defensive Fixes Applied

1. normalize_registration(plate):
   - Bugs Found: Passing None raised AttributeError ('NoneType' has no attribute 'strip'); strings containing only spaces/punctuation ('---', ' ') or non-ASCII characters returned empty or invalid registration plates.
   - Fix Applied: Checked isinstance(plate, str), stripped spaces/dashes/dots, and verified the cleaned plate is non-empty ASCII alphanumeric (cleaned.isascii() and cleaned.isalnum()), returning None for invalid input.

2. total_claim_amount(text):
   - Bugs Found: Passing None raised AttributeError; empty strings (''), trailing/consecutive semicolons ('12000;;800'), non-numeric strings ('abc'), negative values ('-500'), or Unicode digits ('²') caused ValueError or wrong totals.
   - Fix Applied: Checked isinstance(text, str) and non-empty content, stripped each semicolon-separated token, verified token.isascii() and token.isdigit(), and returned None if any token is malformed.

3. driver_age(dob, current_year=2026):
   - Bugs Found: Passing None raised AttributeError; strings with fewer than two '/' characters ('', '14-08-1999') raised IndexError; non-numeric years ('//') raised ValueError; invalid calendar dates ('31/02/1999', '99/99/1999') and future birth years ('01/01/2027' -> age -1) were accepted.
   - Fix Applied: Checked isinstance(dob, str), verified exactly 3 ASCII-digit parts (DD/MM/YYYY with 4-digit year), validated the calendar date via datetime.date(year, month, day), and enforced 0 <= age <= 120.

Reflection Questions

Q1. Why did manual unit testing miss these bugs while Hypothesis found them immediately?
Manual tests usually check expected happy-path inputs (like 'ka 01 ab 1234', '12000;4500;800', and '14/08/1999'). Hypothesis automatically explores edge cases such as None, empty strings (''), delimiter-only strings ('//', ';'), Unicode characters, and out-of-range dates, shrinking any failure to the minimal counterexample.

Q2. Which edge cases were the trickiest to fix properly?
Handling Unicode digits and case conversions (for example, '²'.isdigit() is True in Python, and German 'ß'.upper() expands to 'SS') as well as validating real calendar dates (rejecting '31/02/1999' and future birth years > 2026) required checking .isascii() and validating dates with datetime.date().

Q3. How does defensive programming improve software security and reliability?
By validating input types, formats, character sets, and domain boundaries before processing, functions fail safely (returning None instead of raising unhandled exceptions or returning negative ages/corrupted totals), preventing crashes and injection or DoS vulnerabilities.